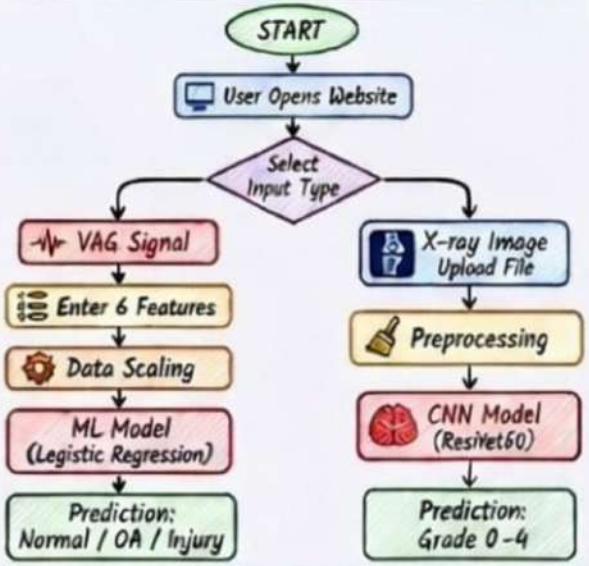
SRU

Multimodal Knee Osteoarthritis Detection Using Vibro-arthrographic Signals and X-ray Images

INTRODUCTION / ABSTRACT

Current OA diagnosis relies heavily on single-modality X-rays, often missing early-stage cases. Vibro-arthrographic (VAG) signals, recording joint sound, offer a non-invasive complementary analysis. This project presents a multimodal approach that integrates both types of data to provide a comprehensive, accurate, and early Knee OA detection system.

SYSTEM ARCHITECTURE



MODEL PERFORMANCE & EVALUATION

- Multi-model evaluation (VAG only)
- Single-model evaluation (X-ray only: EfficientNet, CNN, ResNet50, DenseNet, etc.)
- Analysis of VAG features
- Analysis of X-ray features
- Model training comparison (SVM, RF, Logistic Regression, CNNs)
  - X-ray ResNet50: 93.3%

CONCLUSION / RESULTS

Our system effectively combines VAG signal analysis and CNN-based X-ray grading. Initial evaluation shows promising results, particularly with the ResNet50 for X-ray images, providing detailed severity grades. Integration of VAG and X-ray prediction results is -going for a fused final diagnosis.
Future work will focus on refining decision-level fusion algorithms.

Achieved: Grade Prediction (0-4)

Achieved: Knee Condition Diagnosis (Normal/OA/Injury)

CONCLUSION

METHODOLOGY SUMMARY

The process involves two main data pipelines: VAG signal feature extraction with subsequent Logistic Regression modeling, and X-ray image pre-processing with ResNet50-based CNN modeling. Fusion strategies are being developed to combine the individual predictions into a comprehensive diagnostic output.

CHALLENGES

- Limited and balanced multimodal dataset acquisition
- Integrating disparate data types (signal time-series and image matrices)
- Feature extraction accuracy for VAG signals in complex environments

REFERENCES

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